

Q.1 version control systems.

[A] git and git hub.

git	gitHub.
→ git is software	→ gitHub is service
→ git can installed in local system.	→ gitHub host on web.
→ git track the change and total history of development project in repository.	→ gitHub is the spaces for the git repository.
→ git can free to used.	→ gitHub use free + paid.

git and gitHub can make ecosystem for data analytical and development. like tracking all the history and backup.

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[B] core concepts.

git repository.

→ git repository is the storing area that store all the changes and tracking all the file and project history.

→ in repository every users have completed history so no changes of and data loss or used for the backup as well.

Commit.

→ commit is the pinpoint to store all the changes in the repository so any one who work on the development project can see who can changes, what and when.

Branch.

→ Branch is the independent development line that users can work on it, experiment without the affecting main line of code.

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Q-2

[A] Python and SQL

→ * Python

python is programming languages ~~in~~ used by the data scientists and data analysis.

Python has rich eco-system of library like pandas.

It is also used for the development of software and object oriented programming languages.

→ * SQL

SQL standardised languages for the data merging, manipulating and analytics.

SQL used full for the ETL and ELT process in the database.

SQL used ~~can~~, ~~work~~ and work on many database like mysql and SQLite.

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[B] elements of programming languages.

① syntax.

- syntax is the rules of the programming languages.
- crammer of the languages.

② libraries.

- pre written code in programming languages which is save time and effort.

③ variables.

- store data that have values.

④ conditions.

- statement who controls the flow of code.

important.

Syntax pre define rules so users can know the how to use programming languages.

libraries are very important in programming languages. it save time and effort. users can install and use it that every time no need write code.

when doing data analysis or developing and software conditions are very useful for the control the flow of code.

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[C] Docker.

containers like Docker use the pre defined set of code who allows to build ~~code~~ software and applications.

the main benefits of this types of containers is work with any development environment. mostly ~~App~~ we used ~~or~~ use build on Docker or Docker like containers.

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[D] IDE

IDE is the open sources development platform. that can used and build software or analytical work.

two popular IDEs are. R studio. and VS code

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Q=3

[A] no-code or low-code tools.

~~(A)~~ nsn.

nsn is the automet flow of working task. ~~once~~ one time we mock the flow of any process. after that we tegging the flow it work automatically and give final out out.

~~(B)~~ Power BI

Power BI is the data analyzic dest board.

where we can manging the data processing dest board and then import data and give output. it can feels like using Excel but do more than Excel to. like Automation.

→ X3) Tableau.

Tableau used for the data manipulating meaning and analysis.

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[B] advantages of using no-code/low code.

→ limited advance feature.

Automation and machine learning are advance feature can offer by R and python. traditional tools like Excel are not.

→ Version control system integration

Excel are not easily connected with vast dataset and web services this tools can.

→ Version control system.

in Excel is very hard to. who change, what and when. tools like git build for this.

→ Automation.

no-code/low code tools are automation workflow like clearing data. so data scientist can focus on more ~~important~~ important work.

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→ Excel ~~are~~ not ~~able~~ work on the ~~very~~ big amount of data. it can be risk of data loss but no-code low-code can do.

→ Excel offer limited number of charts while tools like seaborn and D3 have more options.

Q=5

[A] $3 + \sqrt{9} - 2^2 \cdot \frac{4}{3}$

→ Result $\leftarrow 3 + \sqrt{9} - (4) - 2^2 * \frac{4}{3}$.

[B] R code to get current working directory

→ `getwd()`

[C] help documentation. `mean()`

→ `?mean()`

(D) Data frame.

`library("tidyverse")`

`Region <- c("North", "South", "East")`

`Sales <- c("45000", "38000", "52000")`

`df_sales_data <- tibble(Region, Sales)`

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Q:4.

[C] common data issues.

① missing values.

is that cell are empty or null.

ex empty cell, N/A

→ remove or the fix missing data.

② inconsistent data.

Sometime in dataset, data are not consistent or same.

ex DD/mm/yyyy and mm/DD/yyyy

→ much source all the data look like same.

③ duplicate records.

many time data weight more then one time or same data but little bit different.

ex ~~John newyork~~
John ny.

→ fix the duplicate data and reweight the same as original.

④ outliers.

data are compitly different from the rest of them.

ex age 99 or -10

[B]

univariate

→ Univariate data use the one type of data is only one variable like making bar chart.

~~multivariate~~ bivariate.

→ more than two type of data or the variable which is depend on with each other, one can change ~~also~~ the impacted on the other as well. like line chart.

multivariate.

→ more than 3 or more variable in the data. it very complex data set. and it hard to identify the impacted on each other.

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